# Sustainable Freshwater *Aquaculture* for India

A comprehensive guide for over 2 crore small and marginal fish farmers — from hatchery operations and pond management to climate-resilient systems.

**3Cr+**
People dependent on fisheries[1]

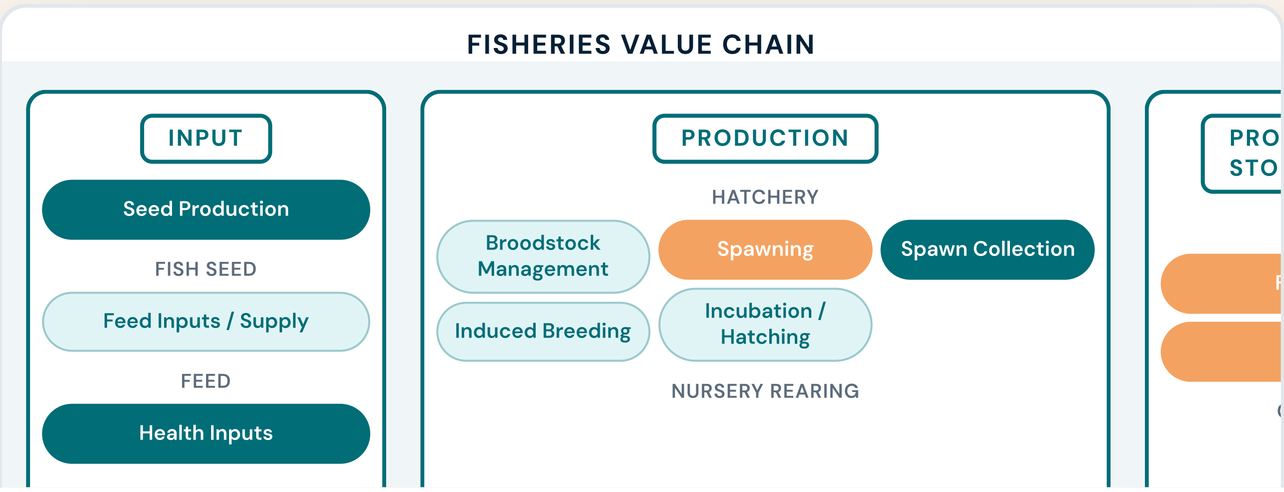
**17MT**
Annual fish production 2022-23[2]

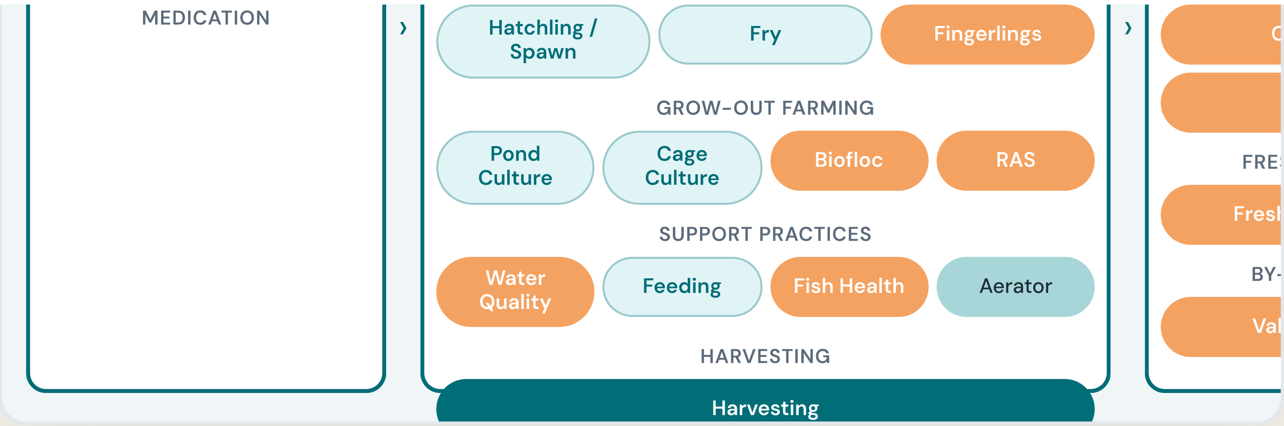
**2nd**
India's global rank in production[3]

[1] PIB India — Fisheries Sector Data [2] Handbook on Fisheries Statistics 2023 (Draft) [3] GroundReport — FAO Ranking

## Fisheries Value Chain

An interactive map of how fish move from seed to market. Click any node to learn what it involves, why it matters, and what challenges and solutions exist at each stage.

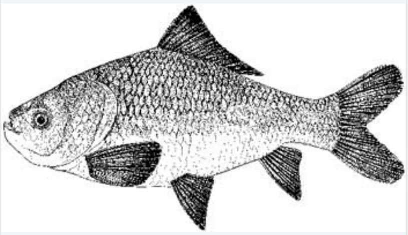





# Species Photo Guide

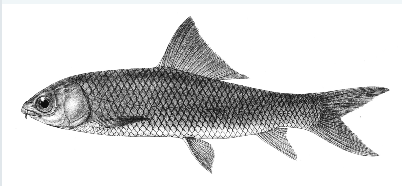
Before choosing what to farm, understand what's available. This chapter introduces the key species — Indian Major Carps, exotic species, high-value indigenous fish, and small indigenous species — with real photographs and guidance on which suits your climate, water type, and market.

All Species | Indian Major Carps | Exotic / Non-Carp | High Value | Small Indigenous



**Catla**
*Catla catla*
IMC

**Rohu**
*Labeo rohita*
IMC



**Mrigal**
*Cirrhinus mrigala*
IMC

### Grass Carp

*Ctenopharyngodon idella*

CARP

### Silver Carp

*Hypophthalmichthys molitrix*

CARP

### Common Carp

*Cyprinus carpio*

CARP

### Amur Carp

*Cyprinus carpio haematopterus*

SELECTIVE BREED

### GIFT Tilapia

*Oreochromis niloticus*

EXOTIC

### Pangasius

*Pangasius hypophthalmus*

EXOTIC

### Magur

*Clarias batrachus*

HIGH VALUE

### Murrel

*Channa striata*

HIGH VALUE

### Pabda

*Ompok bimaculatus*

HIGH VALUE

**Kawai / Climbing Perch**

*Anabas testudineus*

HIGH VALUE

**Singhi**

*Heteropneustes fossilis*

HIGH VALUE

**Mola**

*Amblypharyngodon mola*

SIS

The following sections walk through each stage of the fisheries value chain — from hatchery operations and species selection to grow-out farming, processing, and marketing. Each chapter covers what the stage involves, the challenges farmers face, and the climate-smart solutions being deployed to overcome them. Click on any section in the sidebar or scroll through to explore.

# Hatchery

The foundation of aquaculture. This chapter covers what hatcheries do, the challenges they face in rural India, and the climate-smart infrastructure being deployed to make seed production reliable and year-round.

0 **Spawning, Incubation & Nursery Systems — What, Why, How**

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# Biofloc Technology

A climate-smart grow-out solution. Biofloc eliminates water exchange, reduces feed costs by 30–40%, and turns waste into food — making it ideal for water-scarce and flood-prone regions.

Biofloc Technology — Zero Water Exchange Farming

## Recirculating Aquaculture Systems (RAS)

The most advanced climate-resilient solution. RAS recycles 95–99% of water indoors, enabling year-round production independent of weather, grid power, or land availability.

Recirculating Aquaculture Systems (RAS) — The Most Intensive System

## Water Quality Parameters

The invisible factor that determines everything. Learn the optimal ranges for temperature, oxygen, pH, ammonia, and more — and what to do when parameters go outside safe limits.

| | | | |
|---|---|---|---|
| TEMPERATURE | 25-32°C | Below 20°C → slow growth | Optimal |
| DISSOLVED OXYGEN | > 5 mg/L | Below 3 → stress; below 1 → mortality | Optimal |
| PH LEVEL | 7.0-8.5 | Below 6 or above 9 → stress | Optimal |
| AMMONIA (NH3) | < 0.02 mg/L | Above 0.5 mg/L → toxic | Monitor |
| NITRITE (NO2) | < 0.1 mg/L | Above 1 mg/L → brown blood disease | Monitor |
| TRANSPARENCY | 25-40 cm | Below 15 → bloom; above 60 → low nutrients | Monitor |
| ALKALINITY | 75-200 mg/L | Stabilises pH; lime corrects low levels | |
| H2S | 0 mg/L | Any detectable level = emergency | |

Optimal

Danger

## Disease Checker

Identify what's wrong with your fish. Select the symptoms you observe and get possible diagnoses with treatments — based on the handbook's disease tables. Always consult a fish pathologist for severe outbreaks.

**Select all symptoms you are observing:**

- [ ] Fish surfacing / gasping at surface
- [ ] White or grey spots on body/fins
- [ ] Ulcers or red lesions on skin
- [ ] Fins fraying, rotting or missing
- [ ] Excessive mucus / slimy body
- [ ] Rapid or sudden fish deaths
- [ ] Bloated abdomen / fluid in cavity
- [ ] Fish rubbing against pond sides
- [ ] Refusal to eat / lethargy
- [ ] Whirling or erratic swimming

Get Diagnosis

## Fish Processing, Solar Drying & Marketing

Post-harvest value creation. This chapter covers how fish are preserved, processed, and marketed — with solar-powered solutions that eliminate diesel dependency and extend shelf life from hours to months.

### Fish Processing, Solar Drying & Marketing — From Water to Market

## Climate-Resilient Technologies: Real Deployments

Proof that it works. Real photographs and verified ROI data from field deployments in Assam, Jharkhand, and Odisha — showing how solar-powered systems eliminate grid dependency and transform small-farmer economics.

NURSERY POND (ASSAM)

**3x**

Survival rate — spawn to fingerling — with spiral diffuser aeration

INCOME INCREASE (ASSAM)

**3179%**

Kalong Kapili pilot. Capex Rs 1,90,000. RoI: 2.13

GROW-OUT AERATOR (JHARKHAND)

**46.5%**

Income increase. Survival up 20%. Capex Rs 5,80,000

COST REDUCTION

**20-54%**

Operational cost reduction across all pilot systems

*160W Diaphragm air blower — nursery aeration system*

*Diffuser-type aeration in operation*

*Nursery aeration — Kalong K Assam*

**Why Climate-Resilient?**

In rural India, grid electricity is unreliable — frequent blackouts and voltage fluctuations force farmers to rely on costly diesel generators. Climate-resilient aquaculture systems **eliminate this constraint entirely**, enabling round-the-clock aeration and processing independent of the grid.

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# Farm Record Keeper

Track your farm's performance. Log harvests, revenue, and expenses — all saved locally in your browser. Export to CSV anytime for accounting or government scheme applications.

POND / TANK NO.

e.g. P1

DATE

07/24/2026

HARVEST QUANTITY (KG)

0

SELLING PRICE (RS/KG)

0

TOTAL EXPENSES (RS)

0

SPECIES

Rohu

+ Add Record | Export CSV | Clear All

| Date | Pond | Species | Qty (kg) | Price (Rs/kg) | Revenue (Rs) | Expenses |
|---|---|---|---|---|---|---|
| No records yet. | | | | | | |

## Farm Profitability Calculator

Plan before you invest. Estimate expected yield, revenue, costs, and energy savings based on real pilot data from Assam, Jharkhand, and Odisha — across pond, cage, biofloc, and RAS systems.

POND AREA (HA)

0.5

FARMING SYSTEM

Pond Farming

PRIMARY SPECIES

Rohu / Catla (IMC)

CULTURE PERIOD (MONTHS)

6

SELLING PRICE (RS/KG)

120

SOLAR SYSTEM?

Yes — Climate Resilient

Calculate Expected Profit